

Roll No.

TCE-501

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2023

ENVIRONMENTAL ENGINEERING—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

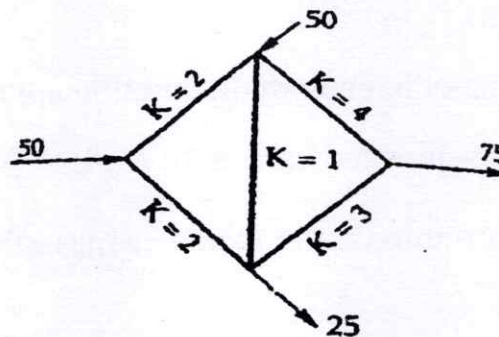
1. (a) Write a short note on the storage zones of a reservoir with the help of a diagram. (CO1)
- (b) Write a short note on dams, their types and selection of site. (CO1)
- (c) Write a short note on provision for "fire demand" in water supply. (CO1)
2. (a) Give the reason for the following : (CO2)
 - (i) hard rubber gasket is used in a flanged joint.
 - (ii) copper pipes are used to convey hot water.
- (b) Write a short note on water hammer pressure and its adverse effects.

P. T. O.

- (c) Assuming the geometric rate of growth of the population of a town, calculate the population of the town in the year 2021 with the help of the following census record of the population :

Year	Population (in thousands)
1991	320
2001	375
2011	440

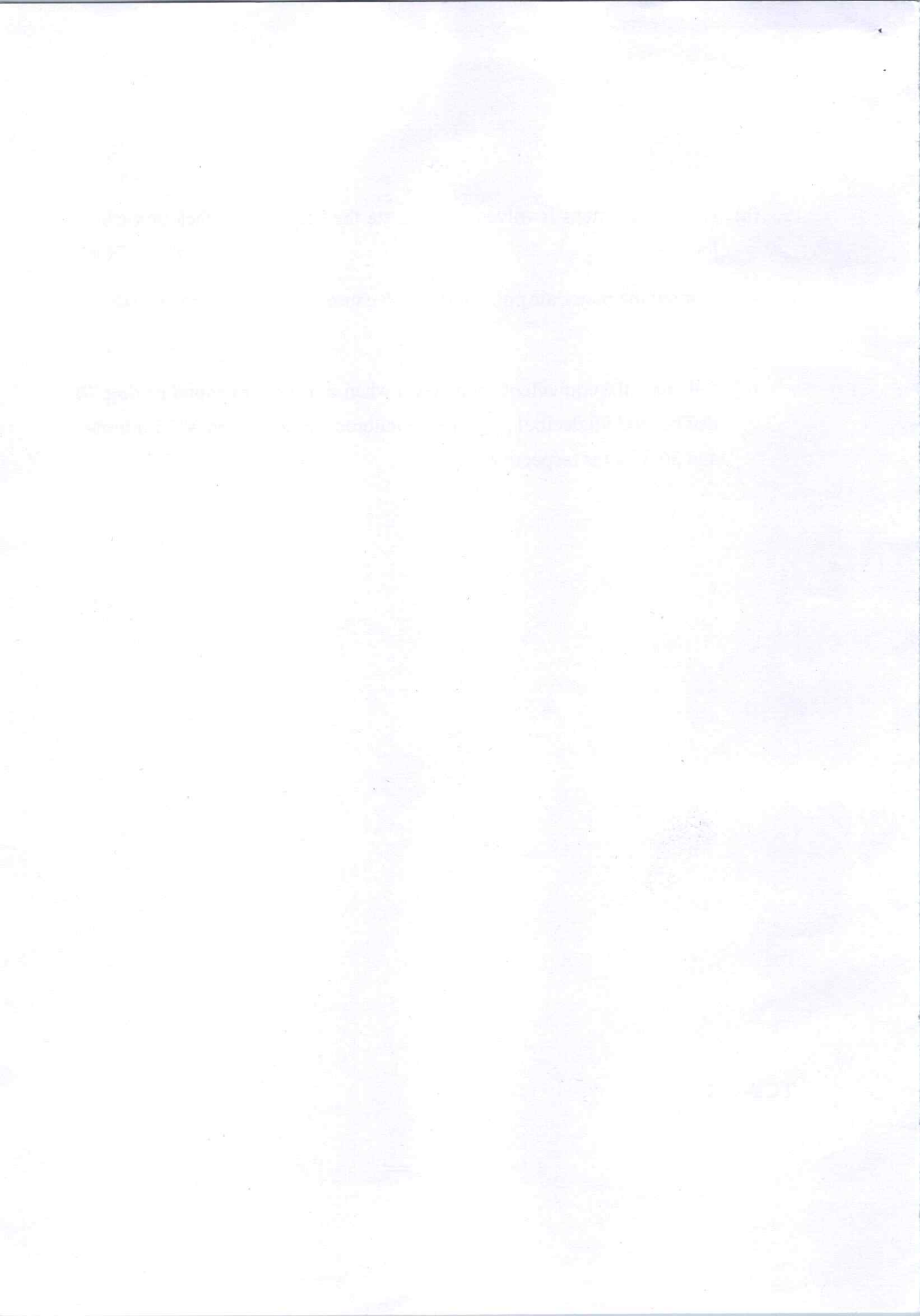
3. (a) Determine the distribution of flow in pipe network as shown in the figure. The head loss at h_L , may be assumed as KQ^n . the flow is turbulent, and pipes are rough. The value of capital K for each pipe is indicated in the figure. use Hardy cross method. The value of n may be assumed as 2. (CO3)



- (b) Describe the function of a distribution reservoir. (CO3)
- (c) Differentiate between the overhead and underground storage. (CO3)
4. (a) Describe the principle of plain sedimentation tank with respect to Stokes' law. (CO4)
- (b) Describe the Lime-soda and Zeolite process with equations. (CO4)
- (c) Differentiate between mechanical and manual screens. (CO4)

(3)

5. (a) List out the steps involved to calculate the height of a stack at various locations. (CO5, CO6)
- (b) List out the major air pollutants which causes impact on human health. (CO5, CO6)
- (c) Calculate the equivalent sound level when 2 sources of sound having 70 decibel and 90 decibel are being monitored for a duration of 25 minutes and 30 minutes respectively. (CO5, CO6)



Roll No.

TCE-502

**B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

REINFORCED CEMENT CONCRETE—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Use of IS 456 : 2000 (Supplied by the Department) is allowed for use in the examination.

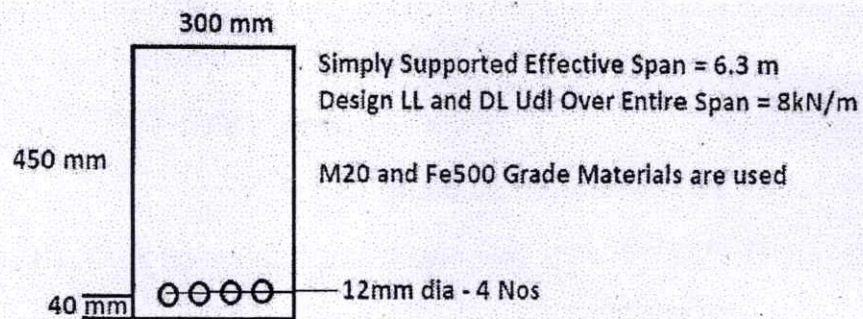
1. (a) Discuss on the various grades of steel used in RCC Design. What is cold working of bars ? (CO1)

(b) Determine the weight proportional ratios of concrete ingredients to produce a M25 Nominal Grade mix of Concrete. Necessary laboratory results of the concrete ingredients are given below : (CO1)

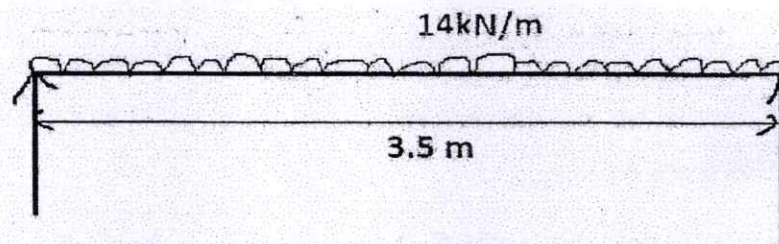
Water-Cement Ratio is 0.45, Void Ratios of Cement, Fine Aggregate, Coarse Aggregate are 50%, 45%, 50% and densities are 1700, 1600, 1700 kg/m³ simultaneously.

P. T. O.

- (c) Explain the concept and feasibility of under reinforced, over reinforced, and balanced sections in the concrete flexure design philosophy. (CO1)
2. (a) Formulate the design constants for a rectangular beam in the Working Stress method of design. Evaluate the design constants, if M20 and Fe415 Gr materials are considered for design. (CO2)
- (b) Determine the Stress in Tension Steel at the given level of a rectangular beam cross section Using WSM. (CO2)



- (c) Design a singly reinforced rectangular section in working stress method of design for the following beam. Use M15 and FE415 Grades of materials.



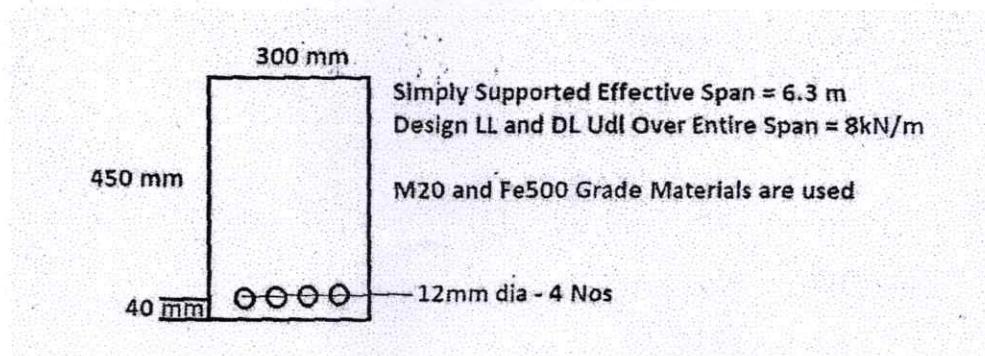
3. (a) Design a T-Beam for a residential structure, in which the beams are laterally spaced at distance of 3.5 m, carrying a slab of thickness 125 mm carrying an overall working live load of 3.5 kN/m² over it. The overall depth of the beam is restricted to 350 mm with a clear web depth

(3)

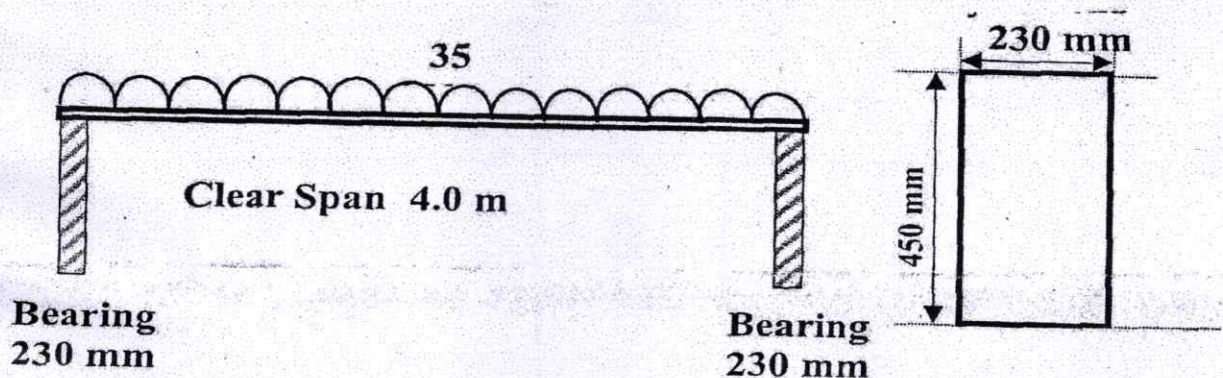
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of 225 mm. M20 and Fe415 grade materials are to be used with a clear top and bottom cover of 15 mm. Assume any other missing data suitably. (CO3)

- (b) Decide the moment carrying capacity of a singly reinforced rectangular section of beam with the given data using limit state method of analysis. M20 and Fe415 grade materials are used. (CO3)

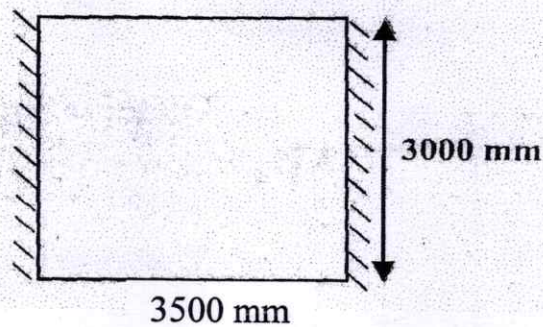


- (c) Design a balanced doubly reinforced rectangular section applying Limit State method with a dimension restriction of 230 mm $\times$ 450 mm, using M20 Grade Concrete and Fe415 Grade Steel material. Top and bottom clear covers are 25 mm. Check for Sufficiency of bond length for the main reinforcement bar. (CO3)

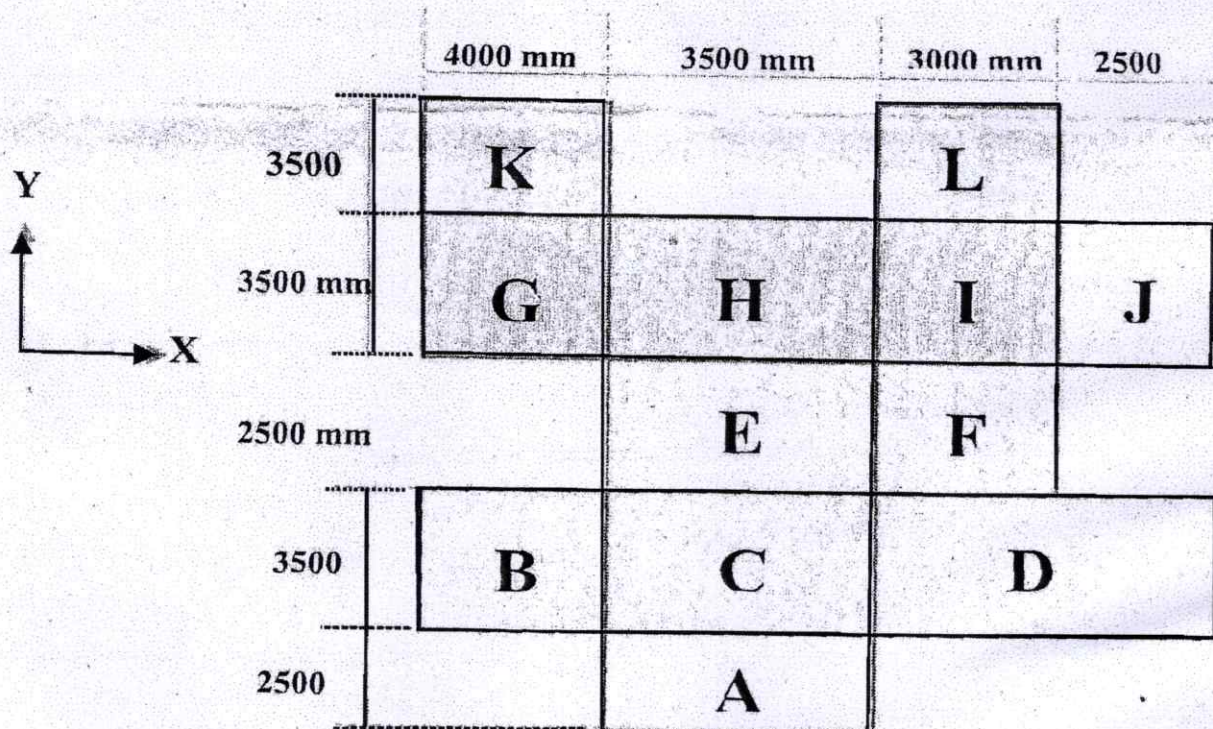


P. T. O.

4. (a) Design and show the rough detailing of reinforcement for the given slab panel. M25 and Fe 415 grade steel is being used. The slab panel is supposed to carry an overall design load of 3.5 kN/m^2 intensity over the entire area. (CO4)



- (b) Decide the design bending moments in X and Y directions of the given slab panel H. (CO4)



- (c) Explain the typical reinforcement detailing and corresponding functional resistance offered by them, for simply supported beam, cantilever beam, one way slab and two way slabs with rough sketches.

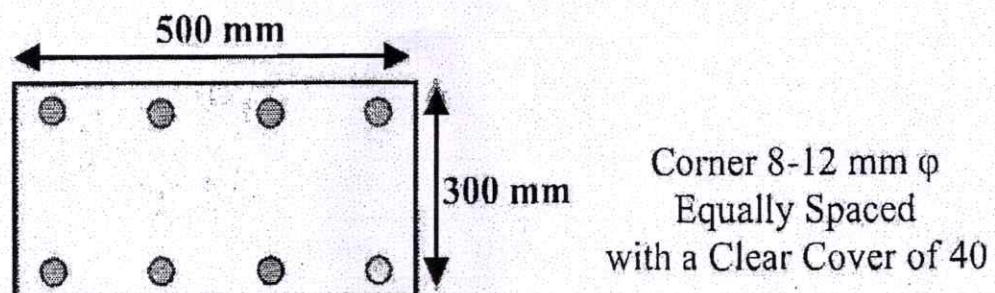
(CO4)

5. (a) Write the general sequential steps of designing a simple footing with equal offsets.

(CO5)

- (b) Determine the Ultimate Load carrying capacity of the given Short Rectangular Column with the given cross section provided with horizontal lateral ties of 6 mm dia bar at a pitch of 160 mm. M 25 and Fe 415 Grade Steels are being used.

(CO5)



- (c) Design reinforcement for a Square Column of size 500×500 mm, subject to an axial load of 2000 kN under service dead and live loads. The column has an unsupported length of 3.0 m and is braced against sideways in both directions. Use M20 and Fe415 Grades of Steel. (CO5)

P. T. O.

Stress-Strain Values for			$F_Y = 500$	
K	Strain	Ultimate Stress-Strain Curve	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	400.0000	347.83	0.001839
0.8500	0.0003	425.0000	369.57	0.002148
0.9000	0.0007	450.0000	391.30	0.002657
0.9500	0.0010	475.0000	413.04	0.003065
0.9750	0.0020	487.5000	423.91	0.004120
1.0000	0.0035	500.0000	434.78	0.005674

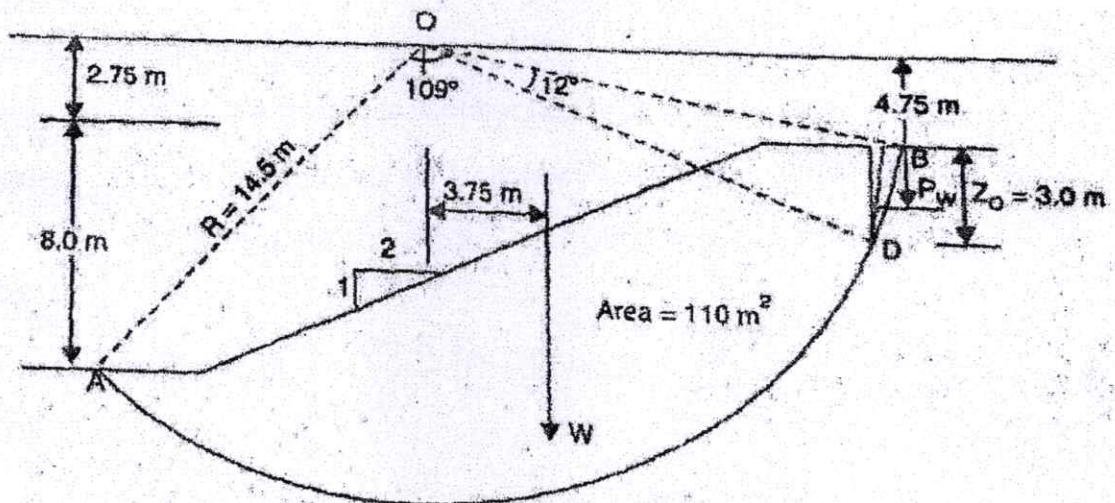
Stress-Strain Values for			$F_Y = 415$	
K	Strain	Ultimate Stress-Strain Curve	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	332.0000	288.70	0.001543
0.8500	0.0003	352.7500	306.74	0.001834
0.9000	0.0007	373.5000	324.78	0.002324
0.9500	0.0010	394.2500	342.83	0.002714
0.9750	0.0020	404.6250	351.85	0.003759
1.0000	0.0035	415.0000	36.87	0.005304

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Time : Three Hours

Note : (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) With reference to the following figure, estimate the factor of safety for the following condition: (i) Tension crack is not developed
(ii) Tension crack is developed and crack is filled with water : (CO1)



P. T. O.

(b) Discuss in detail types of slopes, causes and type of failure of slopes.

(CO1)

(c) Discuss various slope protection measures taken to improve the stability.

(CO1)

2. (a) (i) Provide definitions of ultimate, net and safe bearing capacities, Allowable bearing pressure.

(CO2)

(ii) What are the Terzaghi's bearing capacity equations limitations ?

(b) (i) Discuss the limitations of Plate load test, Standard penetration test and cone penetration test.

(ii) Discuss the effect of ground water table on bearing capacity of shallow foundation.

(CO2)

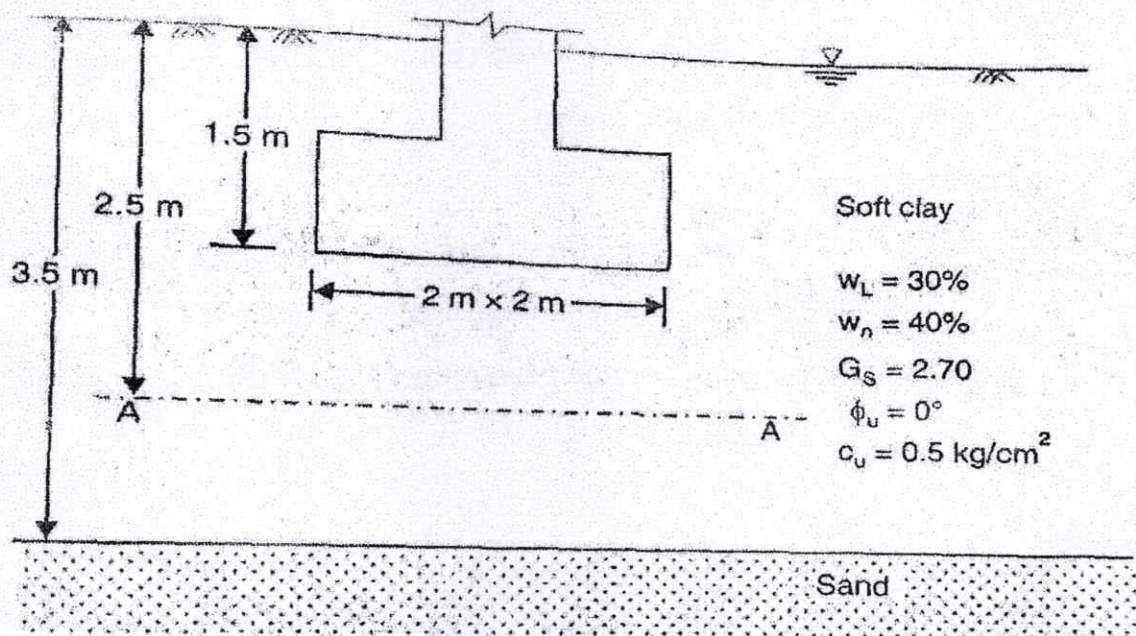
(c) Proportion a square footing to carry a load of 150 ton from a column. The depth of foundation is to be kept at 2 m below the ground surface. Maximum permissible settlement of the footing is 40 mm and a factor of safety of 3.0 is required against shear failure. The subsoil is sand with an average corrected N value of 18 as established from boring. Water Table is at large depth. Use Teng's correlation :

(CO2)

$$q_{np} = 0.14(N-3) \left(\frac{B+0.3}{2B} \right)^2 R'_w C_D S_a$$

where $C_D = 1 + \frac{D}{B} \leq 2.0$.

3. (a) Discuss importance and concept of settlement analysis, i. e. Immediate, Consolidation and Secondary settlements, provide equations for the same and discuss in detail. (CO3)
- (b) Discuss the following : (CO3)
- (i) Esolated, combined, strip and mat foundations
 - (ii) Factors influencing the selection of depth of foundation
 - (iii) Total and differential settlements of footings
 - (iv) Factors influencing the choice of foundation
- (c) Estimate the primary consolidation settlement of the footing placed over normally consolidated soil. Assume the net safe load is applied. Bearing Capacity factor $N_c = 6.9$ and Factor of safety = 3.0. Rest of the geometrical and geotechnical details are provided in the following figure :



4. (a) A pile 45 cm in diameter, 20 m long is driven into a clay soil having following geotechnical properties : $C_u = 45$ kPa, $\phi_u = 0$ and $\gamma_{\text{sat}} = 20$ kN/m³. Ground water is almost at the ground surface. If the submerged weight of the pile is 50 kN, estimate the ultimate uplift capacity of pile under undrained condition. Use $\alpha = 0.95$. (CO4)
- (b) A symmetrical 16 pile group in soft clay, with unconfined compressive strength of 45 kPa, is to be used as foundation for a column. The piles are 350 mm in diameter and 12 m long and spaced at 1000 mm c/c. Determine the maximum load the group can carry with the piles failing
(i) individually (ii) as a block. Assume $\alpha = 1.0$. (CO4)
- (c) Draw typical cross-section of well foundation and indicate different components along with their functions. (CO4)
5. (a) List down the various objectives of sub-surface exploration program. Discuss various methods of sub-surface exploration. (CO5)
- (b) Discuss the following ground improvement techniques : (CO5)
- (i) Chemical stabilization with cement and lime
- (ii) Mechanical stabilization
- (c) Answer the following : (CO5)
- (i) What are the corrections applied to field N-value before they are used in design charts or empirical correlations ?
- (ii) How is the static cone penetration test is different from the standard penetration test ?

Roll No.

TCE-504

**B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

WATER RESOURCES ENGINEERING-I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the various forms of precipitation. (CO1)
(b) Discuss briefly the Horton's method of determining the infiltration capacity of very large basins. (CO1)
(c) The location coordinates in km of five rain gauge stations w.r.t. X are as follows : (CO1)

Station	x and y coordinates (km) w.r.t. X
X	0, 0
A	20, 25
B	-40, 15
C	-30, -20
D	25, -15

P. T. O.

The annual precipitation X for the year 2023 is missing. The annual precipitation at the remaining four stations at X is as follows :

Station	Precipitation (mm)
A	2735
B	2805
C	2680
D	2560

Estimate the missing precipitation at X for the year 2023.

2. (a) What is runoff ? Discuss various factors which affect runoff from a catchment. (CO2)
- (b) Ordinates of 3-hr UH are below, derive the ordinates of 6-hr and plot and same. (CO2)

Time (hr)	3-hr UHGO
0	0
3	1.5
6	4.5
9	8.6
12	12.0
15	9.4
18	4.6
21	2.3
24	0.8
27	0

- (c) Give below are the ordinates of a unit hydrograph for storm of 4-hr duration. Determine the ordinates of food hydrograph when the maximum flood observed was $4000 \text{ m}^2/\text{s}$ and base flow $250 \text{ m}^3/\text{s}$:

(CO2)

Time (hr)	UHGO (in m^3/s)
0	0
4	1500
8	1200
12	600
16	220
20	80
24	0

3. (a) What is meant by artificial recharge of groundwater ? Enumerate the different methods which are used for this purpose and describe any *one* of them briefly. (CO3 and CO6)
- (b) Define duty and delta. Derive an expression establishing relationship between the duty of irrigation water and delta. (CO3 and CO6)
- (c) A well penetrates in to unconfined aquifer having a saturated depth of 50 m. The discharge is 120 litres per minute of 8 m drawdown. Assuming equilibrium flow conditions and homogeneous acquifer, estimate the discharge at 12 m drawdown. Assume radius of influence as same for both the cases. (CO3 and CO6)
4. (a) Explain with the neat sketch how do you carry out lift irrigation scheme. (CO4)

P. T. O.

- (b) What factors you will keep in mind while selecting site for a dam reservoir ? (CO4)
- (c) Following data pertain to monthly flows in river at a particular site :

Month	Flow (in cumecs)
Jan.	65
Feb.	55
March	45
April	30
May	25
June	35
July	60
Aug.	90
Sept.	105
Oct.	100
Nov.	90
Dec.	75

Determine the minimum storage required so as to supply constant discharge of 50 cumecs. (CO4)

5. (a) What are the different types of cross drainage works ? State the conditions under which each one is adopted. Sketch any *one* of them.

(CO5)

- (b) What are the different types of alignment of irrigation canals ? Mention the situation where each type is suitable. (CO5)

- (c) The slope of a channel in alluvium soil is $1/5900$. Find the channel section and maximum discharge which can be allowed to flow in it. Take mean sediment size 0.32 mm. The channel is trapezoidal section having side slope 0.5 : 1.0. (CO5)

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TCE-505

**B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2023**

STRUCTURAL ANALYSIS—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

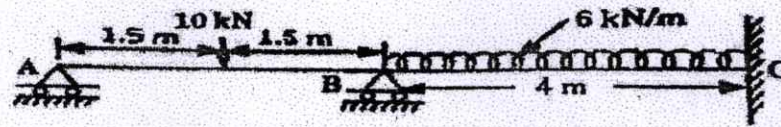
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

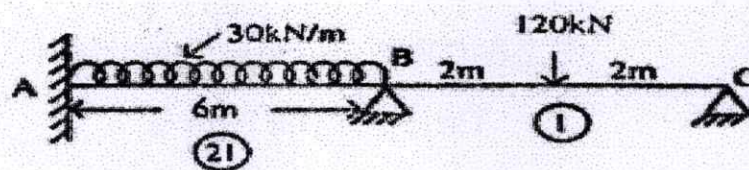
1. (a) A symmetrical two hinged parabolic arch has a span of 40 m and a central rise of 4 m. Find the horizontal thrust at the support if the central 20 m span carries a u.d.l. of 24 kN/m measured horizontally. (CO1)
- (b) Determine the Horizontal thrust in a semi-circular arch of radius R subjected to a concentrated load W at the crown. (CO1)
- (c) A three hinged segmental arch of span 40 m and central rise of 10m is subjected to a UDL of 20 kN/m on its left half and a single concentrated load of 40 KN at 12.5 m from the right support. Find the vertical and horizontal reactions at supports as well as moment at 12.5 m from the left support. (CO1)

P. T. O.

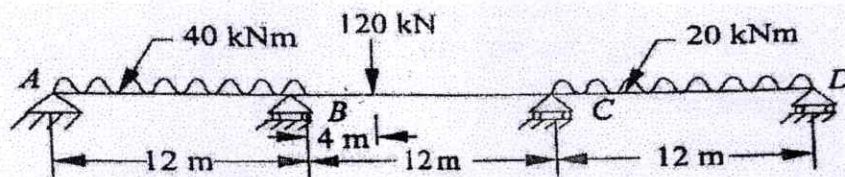
2. (a) Analyse the beam by Flexibility Matrix method and draw BMD : (CO2)



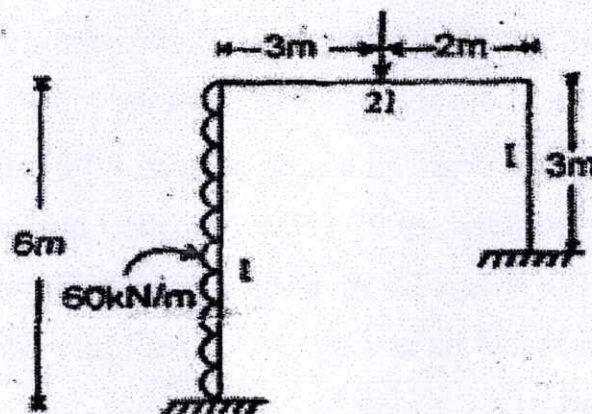
- (b) Analyse the beam by Flexibility Matrix method and draw BMD : (CO2)



- (c) Analyse the beam by Flexibility Matrix method and draw BMD. (CO2)



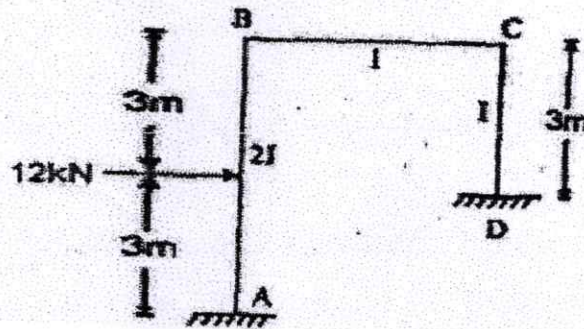
3. (a) Analyse the given frame by the Moment Distribution method. Draw the Bending Moment diagram : (CO3)



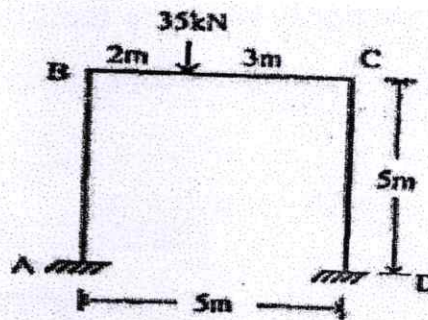
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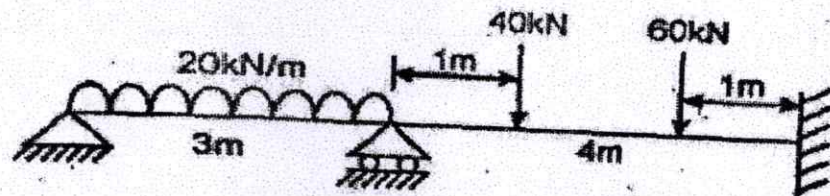
- (b) Analyse the given frame by the Moment Distribution method. Draw the Bending Moment diagram : (CO3)



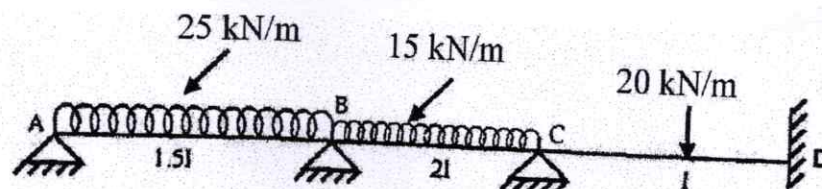
- (c) Analyse the given frame by the Slope Deflection Equation Method and draw the Bending Moment diagram. (CO3)



4. (a) Analyse the continuous beam shown in figure by stiffness matrix method. Take EI is constant. (CO4)

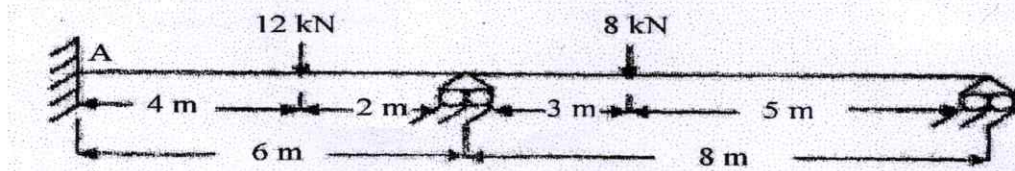


- (b) Analysis the continuous beam shown in figure by stiffness matrix method. Take EI is constant. (CO4)

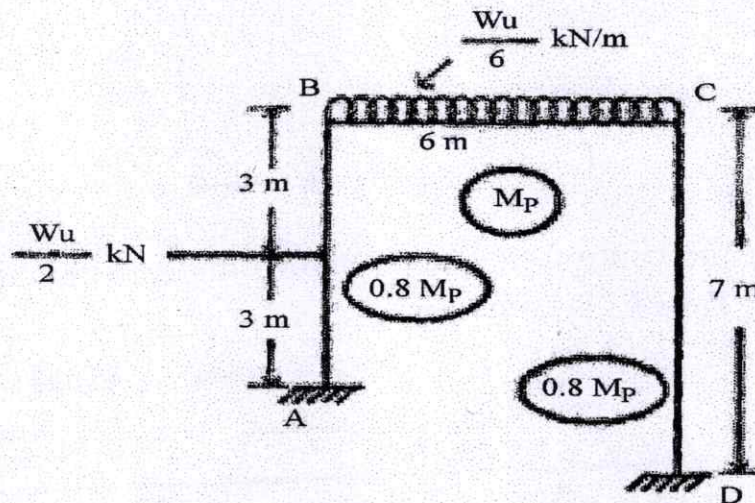


P. T. O.

- (c) Analyse the continuous beam shown in figure by stiffness matrix method. Take EI as constant. (CO4)



5. (a) A propped cantilever beam is loaded with uniformly distributed load (UDL) of intensity 'w' per unit length over its whole span. If the beam has uniformly moment capacity μ_p , locate the plastic hinges and find the collapse load. (CO5)
- (b) Find the collapse load for the loaded frame. The plastic moment capacity for beam is M_p and for the column is $0.80 M_p$: (CO5)



- (c) Calculate the shape factor for a rectangular section. (CO5)